



Tuberculosis

Overview

Tuberculosis, also called TB, is a serious illness that mainly affects the lungs. The germs that cause tuberculosis are a type of bacteria.

Tuberculosis can spread when a person with the illness coughs, sneezes or sings. This can put tiny droplets with the germs into the air. Another person can then breathe in the droplets, and the germs enter the lungs.

Tuberculosis spreads easily where people gather in crowds or where people live in crowded conditions. People with HIV/AIDS and other people with weakened immune systems have a higher risk of catching tuberculosis than people with typical immune systems.

Medicines called antibiotics can treat tuberculosis. But some forms of the bacteria no longer respond well to treatments.

Symptoms

When TB germs survive and multiply in the lungs, it is called a TB infection. A TB infection may be in one of three stages. Symptoms are different in each stage.

Primary TB infection. The first stage is called the primary infection. Immune system cells find and capture the germs. The immune system may completely destroy the germs. But some captured germs may still survive and multiply.

Most people don't have symptoms during a primary infection. Some people may get flu-like symptoms, such as:

- Low fever.
- Tiredness.
- Cough.

Latent TB infection. Primary infection is usually followed by the stage called latent TB infection. Immune system cells build a wall around lung tissue with TB germs. The germs can't do any more harm if the immune system keeps them under control. But the germs survive. There are no symptoms during latent TB infection.

Active TB disease. Active TB disease happens when the immune system can't control an infection. Germs cause disease throughout the lungs or other parts of the body. Active TB disease may happen right after primary infection. But it usually happens after months or years of latent TB infection.

Symptoms of active TB disease in the lungs usually begin gradually and worsen over a few weeks. They may include:

- Cough.
- Coughing up blood or mucus.
- Chest pain.
- Pain with breathing or coughing.
- Fever.
- Chills.
- Night sweats.
- Weight loss.
- Not wanting to eat.

- Tiredness.
- Not feeling well in general.

Active TB disease outside the lungs. TB infection can spread from the lungs to other parts of the body. This is called extrapulmonary tuberculosis. Symptoms vary depending on what part of the body is infected. Common symptoms may include:

- Fever.
- Chills.
- Night sweats.
- Weight loss.
- Not wanting to eat.
- Tiredness.
- Not feeling well in general.
- Pain near the site of infection.

Active TB disease in the voice box is outside the lungs, but it has symptoms more like disease in the lungs.

Common sites of active TB disease outside the lungs include:

- Kidneys.
- Liver.
- Fluid surrounding the brain and spinal cord.
- Heart muscles.
- Genitals.
- Lymph nodes.
- Bones and joints.

- Skin.
- Walls of blood vessels.
- Voice box, also called larynx.

Active TB disease in children. Symptoms of active TB disease in children vary. Typically, symptoms by age may include the following:

- **Teenagers.** Symptoms are similar to adult symptoms.
- **1- to 12-year-olds.** Younger children may have a fever that won't go away and weight loss.
- **Infants.** The baby doesn't grow or gain weight as expected. Also, a baby may have symptoms from swelling in the fluid around the brain or spinal cord, including:
 - Being sluggish or not active.
 - Unusually fussy.
 - Vomiting.
 - Poor feeding.
 - Bulging soft spot on the head.
 - Poor reflexes.

When to see a doctor

The symptoms of tuberculosis are similar to symptoms of many different illnesses. See your healthcare professional if you have symptoms that don't improve with a few days of rest.

Get emergency care if you have:

- Chest pain.
- Sudden, severe headache.
- Confusion.
- Seizures.
- Difficulty breathing.

Get immediate or urgent care if you:

- Cough up blood.
- Have blood in your urine or stool.

Causes

Tuberculosis is caused by a bacterium called *Mycobacterium tuberculosis*.

People with active TB disease in the lungs or voice box can spread the disease. They release tiny droplets that carry the bacteria through the air. This can happen when they're speaking, singing, laughing, coughing or sneezing. A person can get an infection after inhaling the droplets.

The disease is more likely to spread when people spend a lot of time together in an indoor space. So the disease spreads easily in places where people live or work together for long periods. Also, the disease spreads more easily in crowded gatherings.

A person with a latent TB infection cannot pass the disease to other people. A person taking medicine to treat active TB disease usually can't pass the disease after 2 to 3 weeks of treatment.

Drug-resistant TB

Some forms of the TB bacteria have become drug resistant. This means that medicines that once cured the disease no longer work.

This happens, in part, because of naturally occurring genetic changes in bacteria. A random genetic change in a bacterium might give it some quality that makes it more likely to survive the attack of an antibiotic. If it does survive, then it can multiply.

When antibiotic medicines aren't used correctly — or medicines fail to clear out all the bacteria for another reason — the conditions are ideal for more-resistant versions of the bacteria to take hold and multiply. If these bacteria are passed on to other people, a new drug-resistant strain can grow over time.

Problems that can lead to such drug-resistant strains of bacteria include the following:

- People didn't follow directions for taking the medicines or stopped taking the medicines.
- They weren't prescribed the right treatment plan.
- Medicines were not available.
- The medicines were of poor quality.
- The body didn't absorb the medicines as expected.

Risk factors

Anyone can get tuberculosis, but certain factors increase the risk of getting an infection. Other factors increase the risk of an infection becoming active TB disease.

The Centers for Disease Control and Prevention recommends a TB test for people who have an increased risk of TB infection or active TB disease. Talk to your healthcare professional if you have one or more of the following risk factors.

Risk of TB infection

Certain living or working conditions make it easier for the disease to pass from one person to another. These conditions increase the risk of getting a TB infection:

- Living with someone with active TB disease.
- Living or traveling in a country where TB is common, including several countries in Latin America, Africa, Asia and the Pacific Islands.
- Living or working in places where people live close together, such as prisons, nursing homes and shelters for homeless people.
- Living in a community identified as being at high risk of tuberculosis.
- Working in healthcare and treating people with a high risk of TB.

Risk of active TB disease

A weakened immune system increases the risk of a TB infection becoming active TB disease. Conditions or treatments that weaken the immune system include:

- HIV/AIDS.
- Diabetes.
- Severe kidney disease.
- Cancers of the head, neck and blood.
- Malnutrition or low body weight.
- Cancer treatment, such as chemotherapy.
- Medicine to prevent rejection of transplanted organs.
- Long-term use of prescription steroids.
- Use of unlawful injected drugs.

- Misuse of alcohol.
- Smoking and using other tobacco products.

Age and active TB disease

The risk of a TB infection becoming active TB disease changes with age.

- **Under 5 years of age.** Until children reach age 5, they have high risk of a TB infection becoming active TB disease. The risk is greater for children under age 2. Tuberculosis in this age group often leads to serious disease in the fluid surrounding the brain and spinal column, called meningitis.
- **Age 15 to 25.** People in this age group have an increased risk of developing more-severe active TB disease in the lungs.
- **Age 65 and older.** The immune system weakens during older age. Older adults have a greater risk of active TB disease. Also, the disease may be more difficult to treat.

Prevention

If you test positive for latent TB infection, you may need to take medicines to prevent active TB disease.

Preventing the spread of disease

If you have active TB disease, you'll need to take steps to prevent other people from getting an infection. You will take medicines for four, six or nine months. Take all of the medicines as directed during the entire time.

During the first 2 to 3 weeks, you will be able to pass TB bacteria to others. Protect others with these steps:

- **Stay home.** Don't go to work or school.
- **Isolate at home.** Spend as little time as possible among members of your household. Sleep in a separate room.
- **Ventilate the room.** Tuberculosis germs spread more easily in small, closed spaces. If it's not too cold outdoors, open the windows. Use a fan to blow air out. If you have more than one window, use one fan to blow air out and another to blow air in.
- **Wear face masks.** Wear a mask when you have to be around other people. Ask other members of the household to wear masks to protect themselves.
- **Cover your mouth.** Use a tissue to cover your mouth anytime you sneeze or cough. Put the dirty tissue in a bag, seal it and throw it away.

Vaccinations

In countries where tuberculosis is common, infants often are vaccinated with the bacille Calmette-Guerin (BCG) vaccine. This protects infants and toddlers who are more likely to have active TB disease in the fluid surrounding the brain and spinal cord.

The vaccine may not protect against disease in the lungs, which is more likely in the United States. Dozens of new TB vaccines are in various stages of development and testing.

By Mayo Clinic Staff

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